

Monomial $f(q) = a \cdot q^4$ Report

Polynomial

$$f(q) = aq^4$$

Naive

$$g_{\text{naive}}(X) = X^4 a$$

$$\mathbb{E}[g_{\text{naive}}(X)] = 24a \frac{\Delta^4}{\epsilon} + 12a \frac{\Delta^2}{\epsilon} q^2 + aq^4$$

$$\text{Var}(g_{\text{naive}}(X)) = 39744a^2 \frac{\Delta^8}{\epsilon} + 19584a^2 \frac{\Delta^6}{\epsilon} q^2 + 1488a^2 \frac{\Delta^4}{\epsilon} q^4 + 32a^2 \frac{\Delta^2}{\epsilon} q^6$$

$$\text{MSE}(g_{\text{naive}}) = 40320a^2 \frac{\Delta^8}{\epsilon} + 20160a^2 \frac{\Delta^6}{\epsilon} q^2 + 1632a^2 \frac{\Delta^4}{\epsilon} q^4 + 32a^2 \frac{\Delta^2}{\epsilon} q^6$$

$$\text{Bias}(g_{\text{naive}}) = 24a \frac{\Delta^4}{\epsilon} + 12a \frac{\Delta^2}{\epsilon} q^2$$

Unbiased

$$g_{\text{unb}}(X) = X^4 a - 12X^2 a \frac{\Delta^2}{\epsilon}$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^4$$

$$\text{Var}(g_{\text{unb}}(X)) = 26496a^2 \frac{\Delta^8}{\epsilon} + 13248a^2 \frac{\Delta^6}{\epsilon} q^2 + 1104a^2 \frac{\Delta^4}{\epsilon} q^4 + 32a^2 \frac{\Delta^2}{\epsilon} q^6$$

$$\text{MSE}(g_{\text{unb}}) = 26496a^2 \frac{\Delta^8}{\epsilon} + 13248a^2 \frac{\Delta^6}{\epsilon} q^2 + 1104a^2 \frac{\Delta^4}{\epsilon} q^4 + 32a^2 \frac{\Delta^2}{\epsilon} q^6$$

Comparison

$$\text{mean gap} = -24a\frac{\Delta^4}{\epsilon} - 12a\frac{\Delta^2}{\epsilon}q^2$$

$$\text{variance gap} = -13248a^2\frac{\Delta^8}{\epsilon} - 6336a^2\frac{\Delta^6}{\epsilon}q^2 - 384a^2\frac{\Delta^4}{\epsilon}q^4$$

$$\text{MSE gap} = -13824a^2\frac{\Delta^8}{\epsilon} - 6912a^2\frac{\Delta^6}{\epsilon}q^2 - 528a^2\frac{\Delta^4}{\epsilon}q^4$$

$$\text{Bias}(g_{\text{naive}})^2 = 576a^2\frac{\Delta^8}{\epsilon} + 576a^2\frac{\Delta^6}{\epsilon}q^2 + 144a^2\frac{\Delta^4}{\epsilon}q^4$$